Measurement and State Collapse

Why Measurement Is Special

Imagine watching a movie. Before pressing Pause, the movie is continuously changing. The moment you press Pause, one frame is frozen on the screen. Quantum measurement behaves in a somewhat similar way. Before measurement, a qubit can exist in a quantum state. After measurement, you observe exactly one classical result. This process is called quantum measurement.

Measurement is the process of reading information from a qubit. Unlike a classical bit, a qubit cannot simply be looked at without affecting its state. When measured, the qubit produces one of the computational basis states, |0> or |1>. No other outcome is possible for a single qubit measured in the computational, or Z, basis.

1. Before vs After Measurement

Suppose a qubit is prepared as |psi> = alpha|0> + beta|1>. Before measurement it is in superposition. After measurement, it becomes |0> or |1>, and the superposition is no longer available.

Analogy: A Multiple-Choice Exam

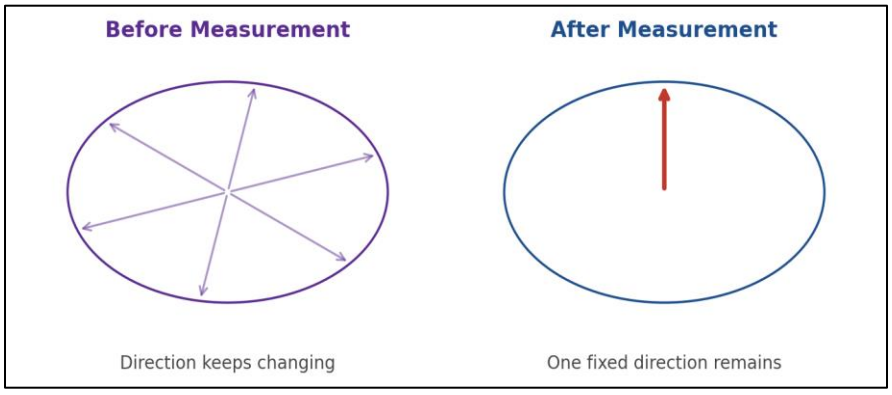
Imagine a student has not yet submitted an online quiz; the answers are still editable, like draft answers. Clicking Submit fixes them into final answers. Similarly, before measurement a qubit is in a quantum state, and measurement produces one definite outcome.

Does Measurement Always Give the Same Result

No. The result depends on the probability amplitudes. For |psi> = (|0> + |1>) / root 2, P(|0>) = 50% and P(|1>) = 50%. If you prepare and measure this state many times, outcomes vary run to run, but over many measurements the results approach the predicted probabilities, similar to flipping a fair coin many times.

2. What Is State Collapse

The transition from a superposition to a definite measured state is called state collapse. Before measurement, a qubit spins through many possible directions; the instant it is measured, one fixed direction remains.



Measuring Again

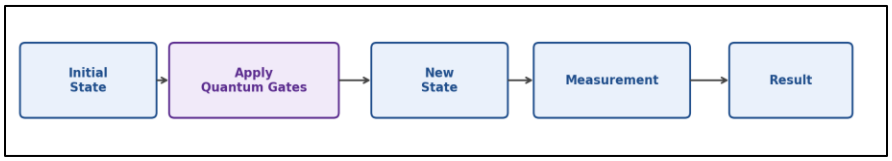
Suppose the first measurement gives |0>. Measuring again immediately gives |0> again, and again, because the state has already collapsed to |0>. There is no remaining superposition to produce a different result.

Measurement Is Irreversible

Think about bursting a balloon: before, a balloon; after, pieces of rubber, and you cannot return to the original balloon by looking at the pieces. Similarly, once a quantum state has been measured, the original superposition cannot be recovered from that measurement alone.

3. Why We Measure at the End

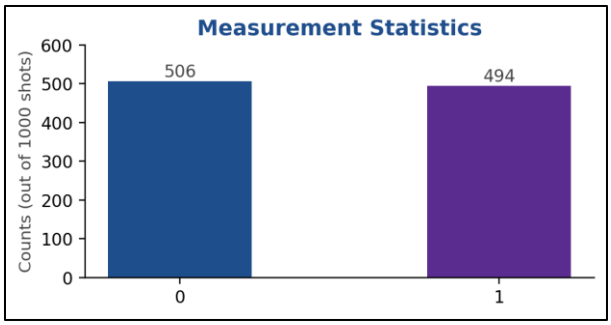
Quantum algorithms manipulate quantum states using quantum gates. Only after all computations are complete do we measure.



If measurement is performed too early, the quantum state collapses before the computation is finished. On IBM Quantum hardware, measurement is represented by the Measurement Gate, and every circuit execution ends with measurement because classical computers can only process classical outputs.

Measurement Statistics

Suppose we execute the same quantum circuit 1,000 times. Instead of looking at a single measurement, quantum programmers analyse many measurements, called shots. More shots generally give a better estimate of the underlying probabilities.



4. Common Beginner Misconceptions

- Measurement reveals the hidden value of the qubit. In reality, the outcome is governed by the state's probability amplitudes, not a hidden fixed value
- Measurement destroys the qubit. In reality, the qubit still exists; it is simply now in the measured basis state

- Measuring twice gives different answers. In reality, if nothing changes between measurements, repeated measurements give the same result because the state has already collapsed

Quick Comparison

Concept	Meaning
Measurement	Reading information from a qubit
Superposition	Multiple basis states combined into one quantum state
Probability	Likelihood of each measurement outcome
State collapse	Transition from a quantum state to one definite basis state
Shots	Repeating the same circuit many times to estimate probabilities
Measurement basis	The set of states used to measure, commonly the Z basis

Summary: What to Remember

- Measurement converts a quantum state into exactly one classical result, |0> or |1>
- A single measurement is probabilistic, governed by the state's probability amplitudes
- State collapse is the transition from superposition to one definite outcome
- Measuring an already-collapsed qubit again gives the same result
- Measurement is irreversible; the original superposition cannot be recovered
- Quantum algorithms measure only at the end, after all gates have been applied
- Shots are repeated executions used to estimate the underlying probability distribution